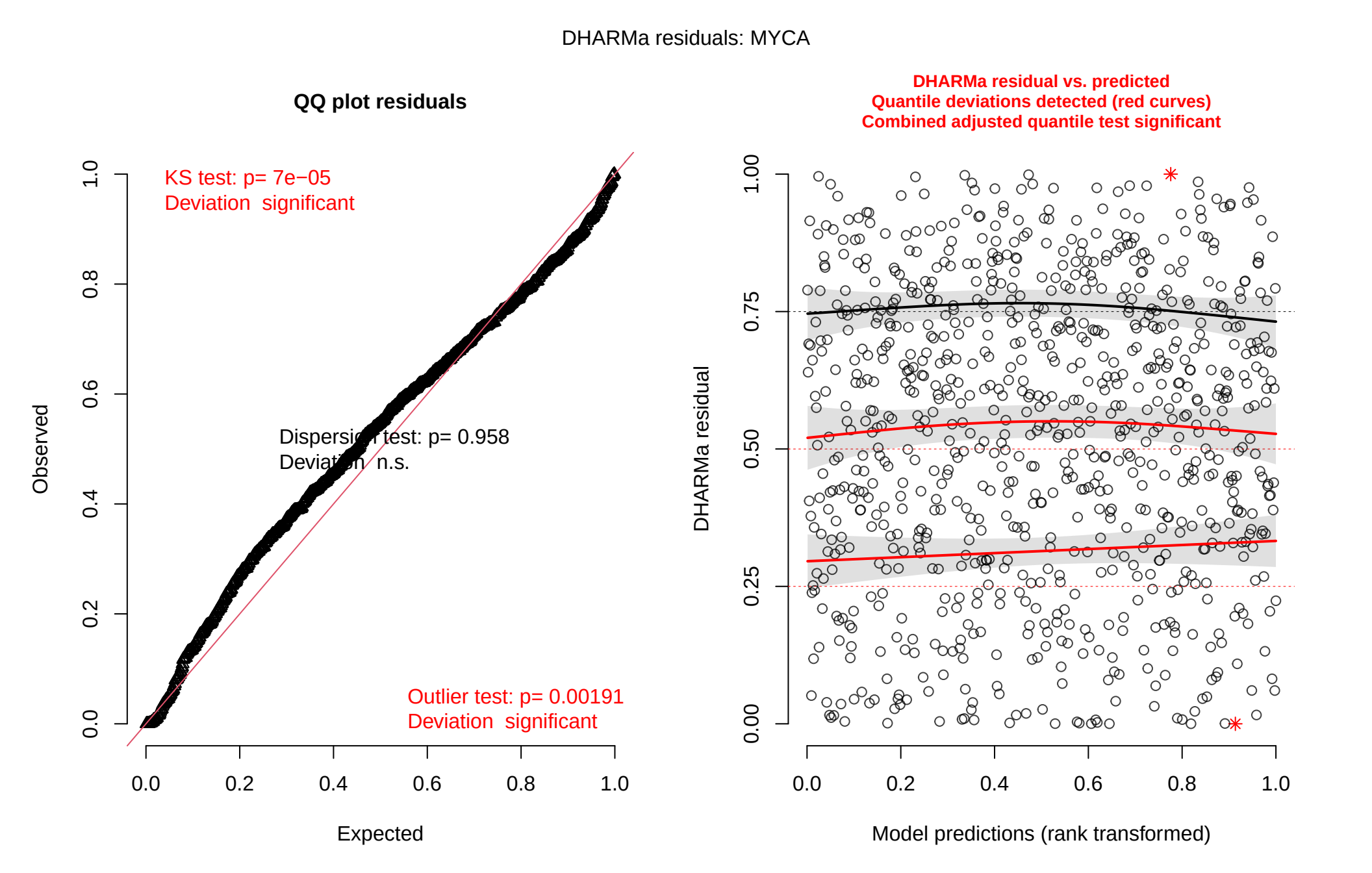
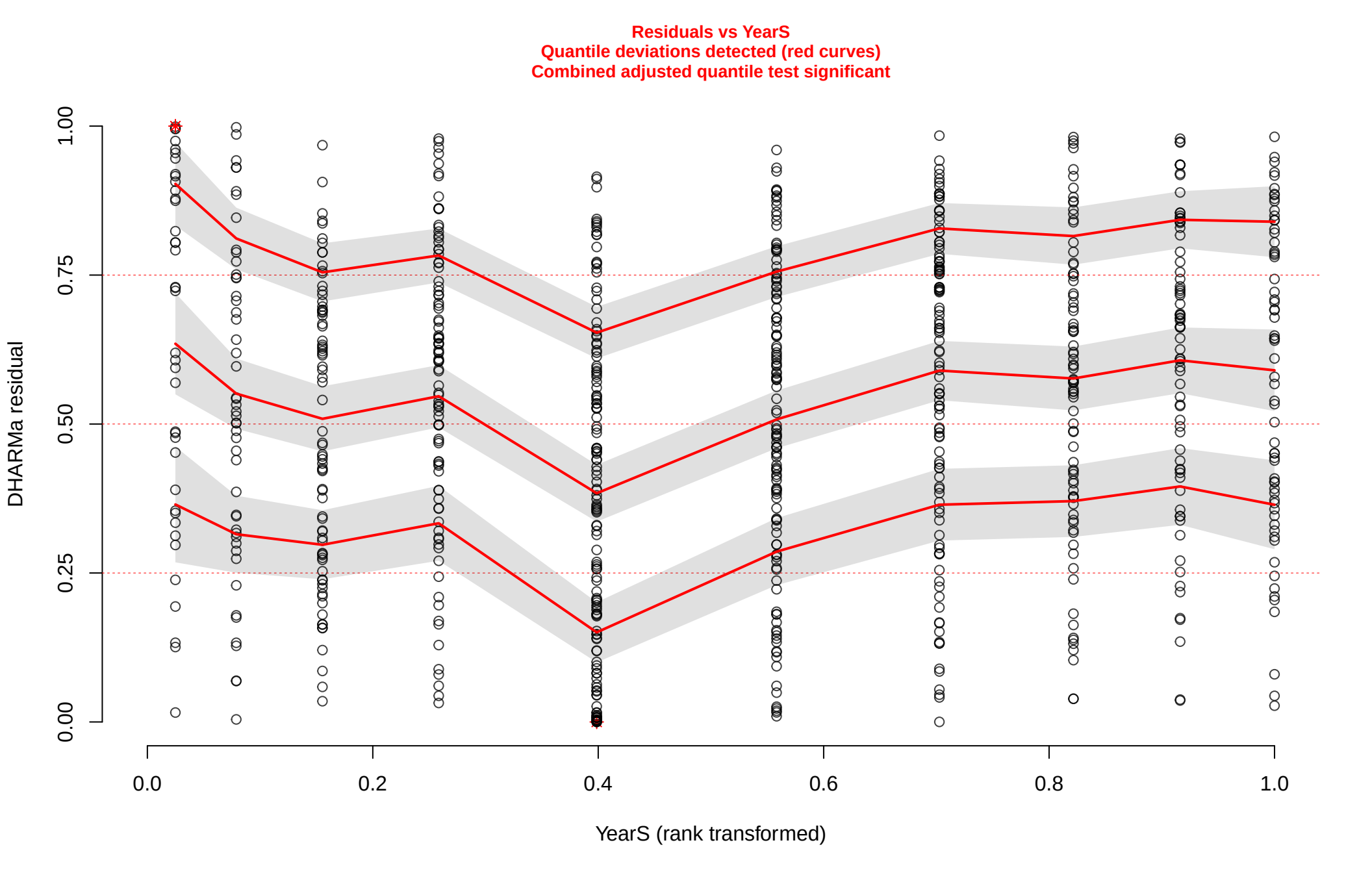
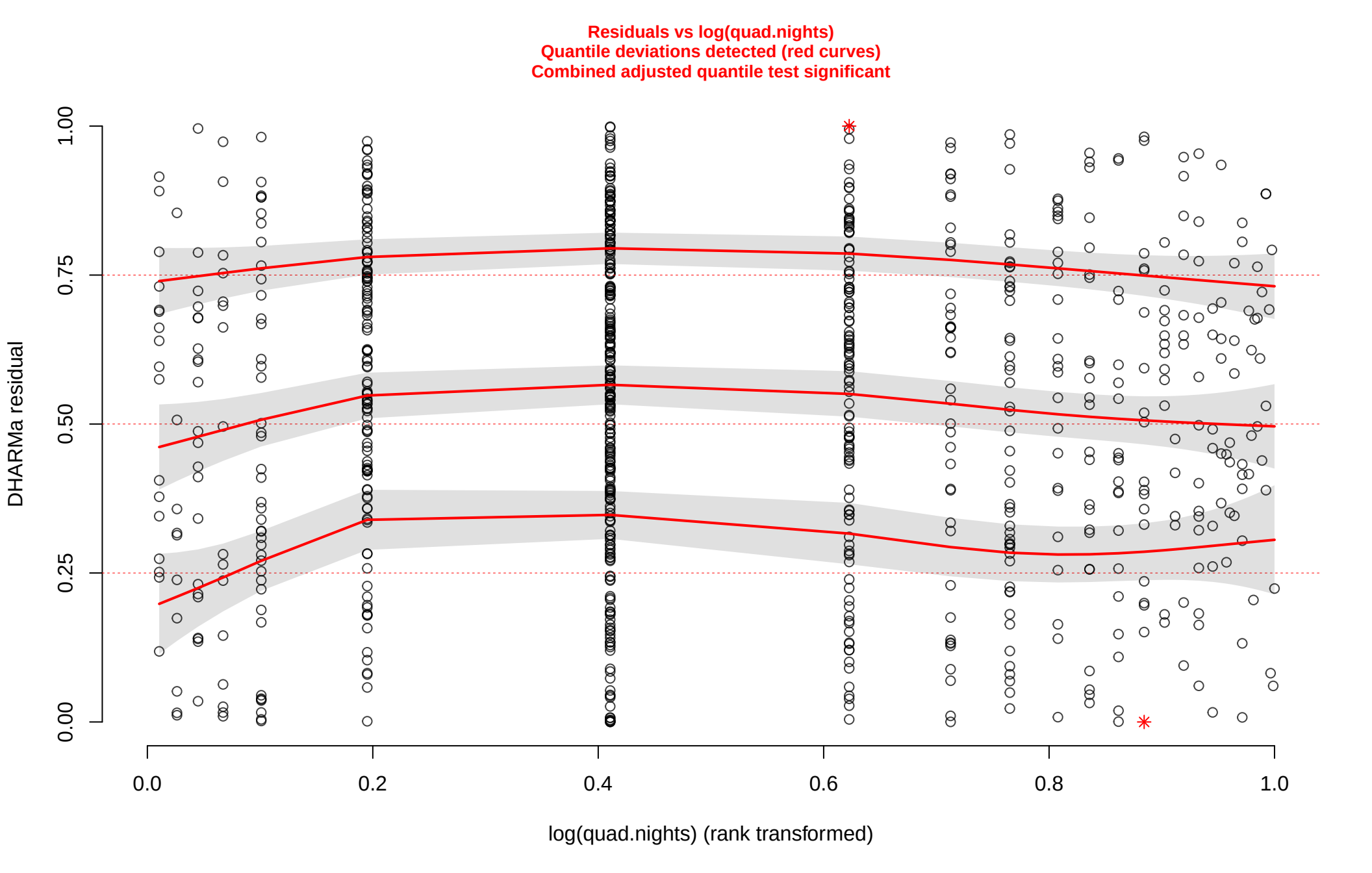
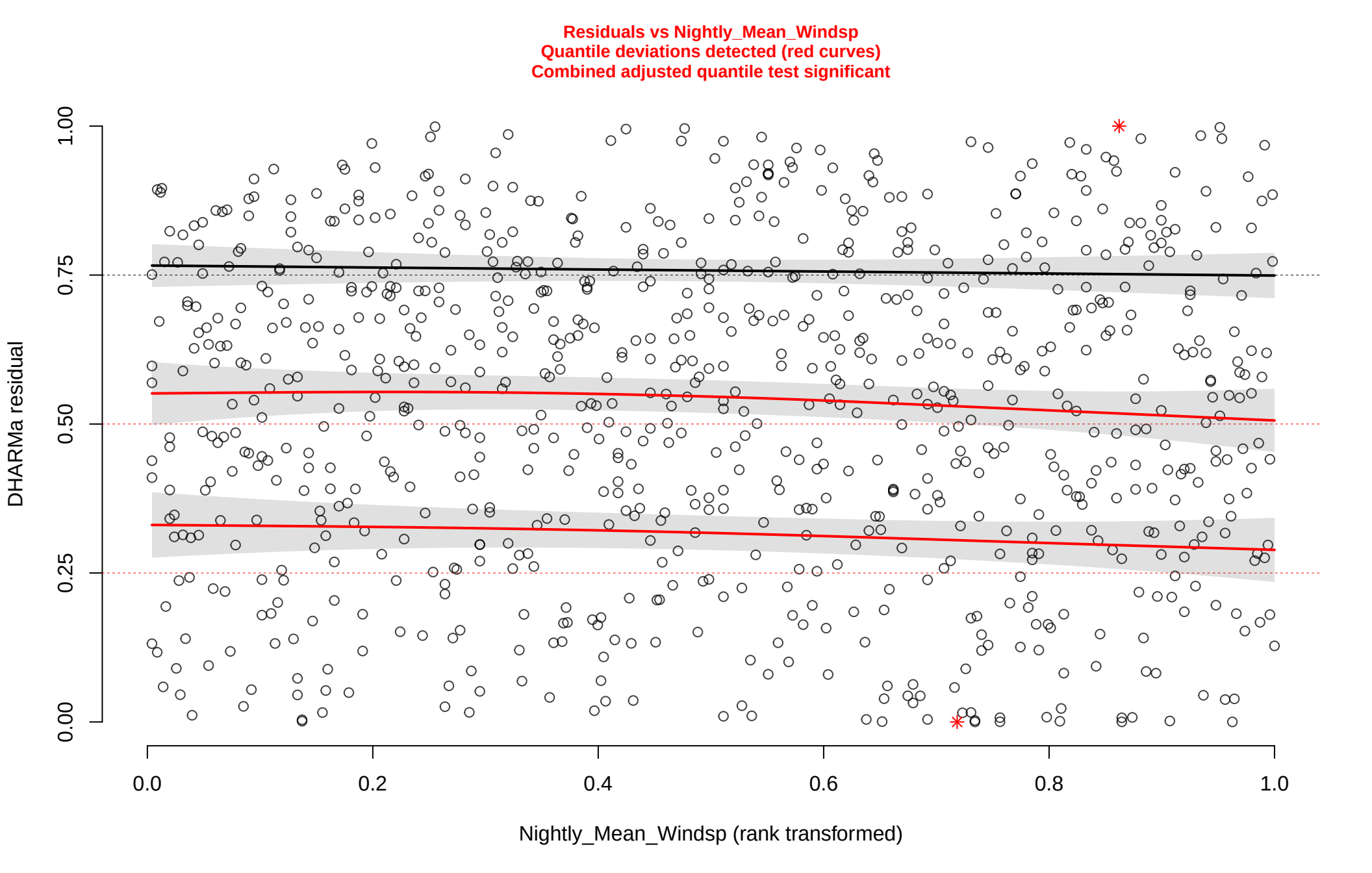
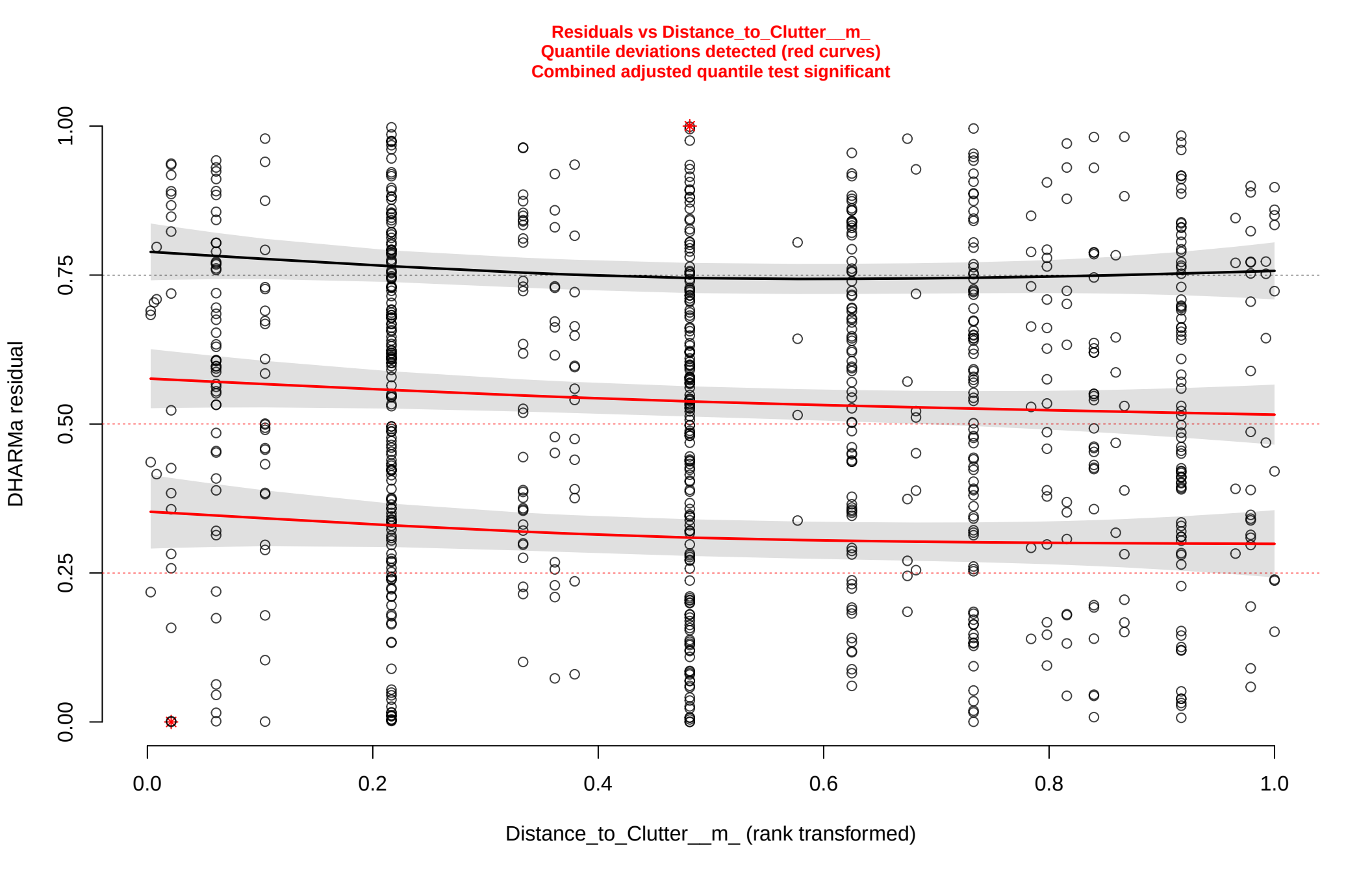
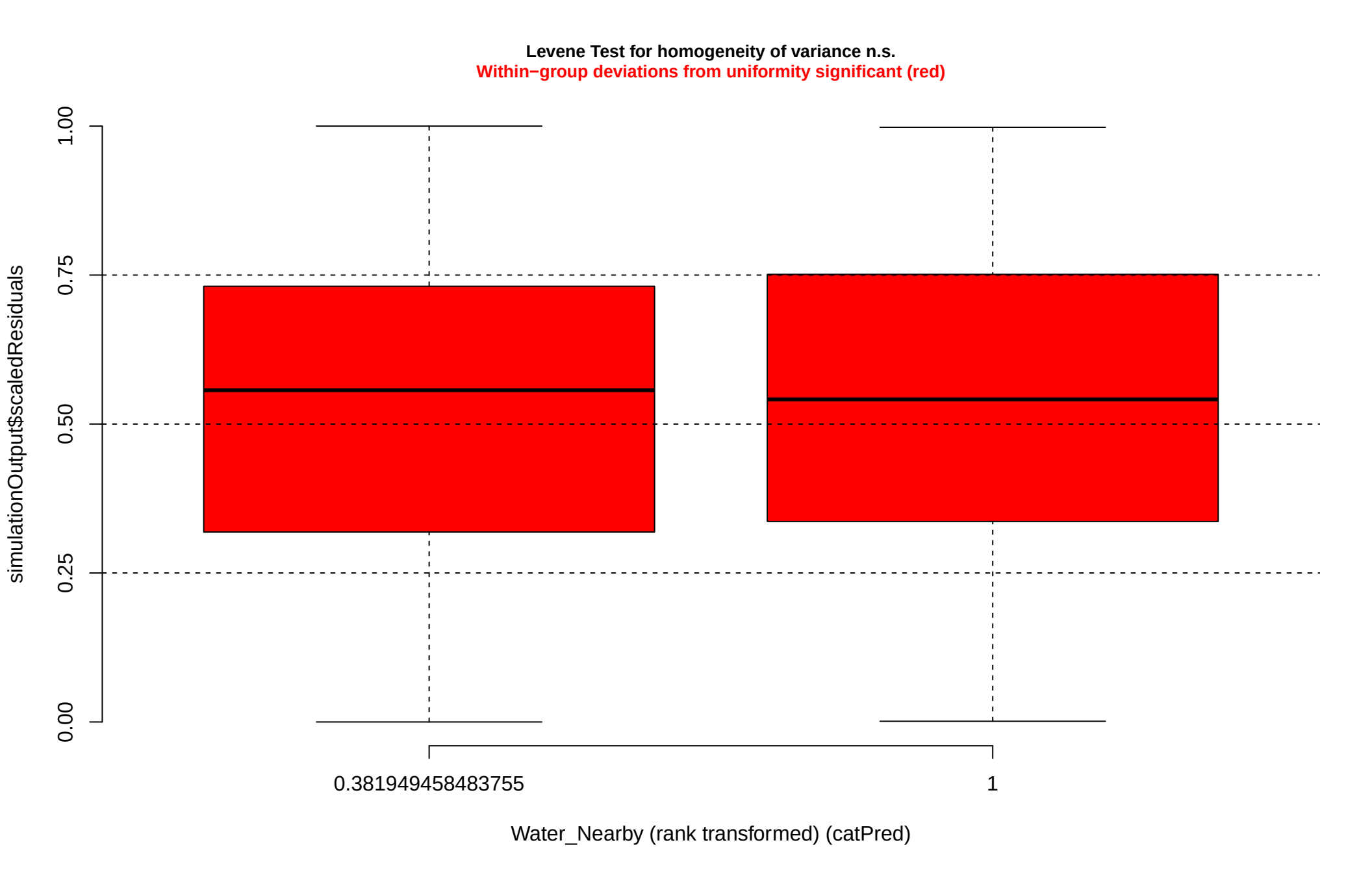

DHARMa residuals: MYCA
QQ plot residuals
KS test: p= 7e−05
Deviation significant
Dispersion test: p= 0.958
Deviation n.s.
Outlier test: p= 0.00191
Deviation significant
Observed
Expected
0.0
0.2
0.4
0.6
0.8
1.0
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
Model predictions (rank transformed)
0.00
0.25
0.50
0.75
1.00

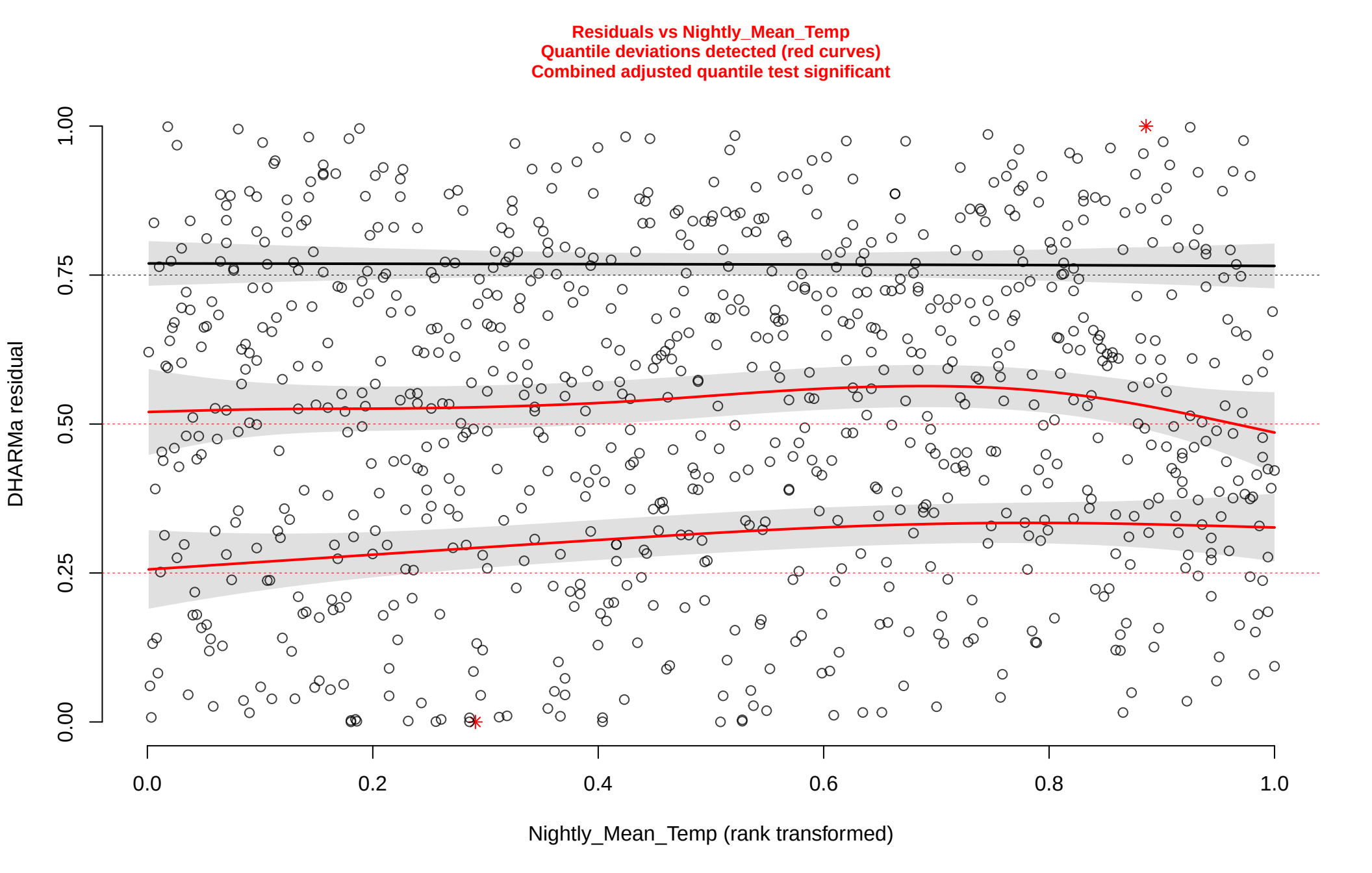

Residuals vs YearS
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

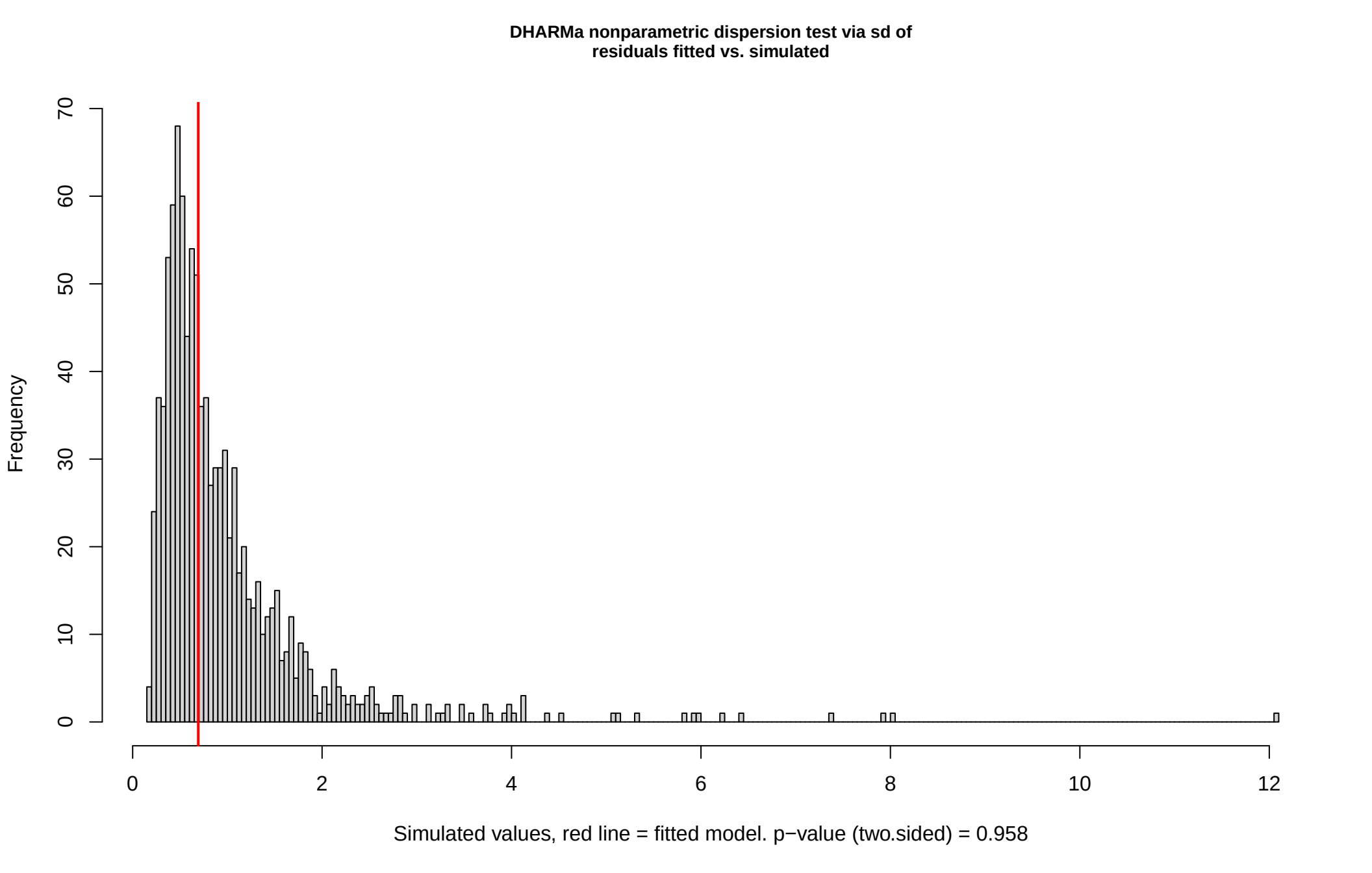

Residuals vs log(quad.nights)
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

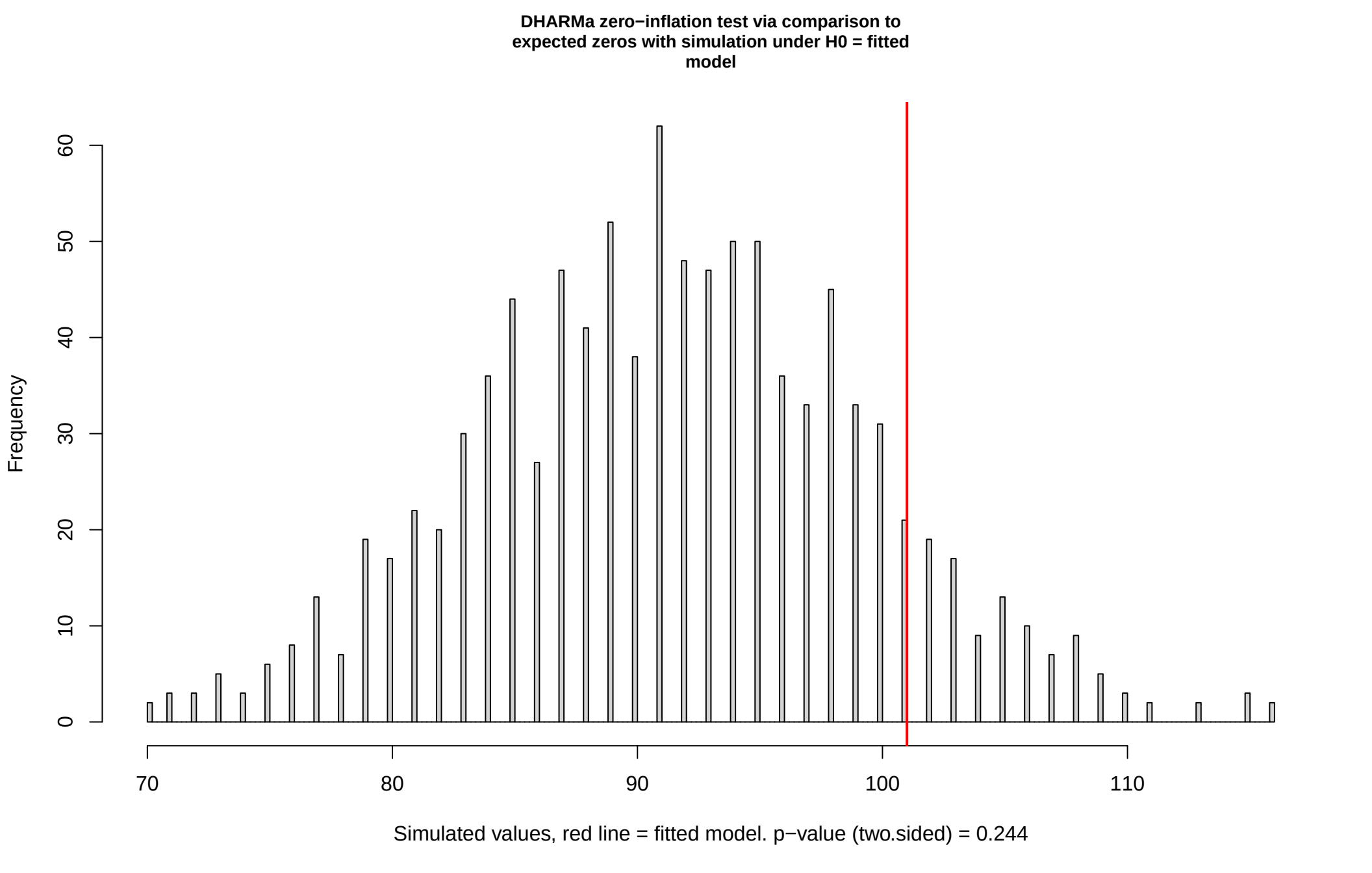

Residuals vs Nightly_Mean_Windsp
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Windsp (rank transformed)

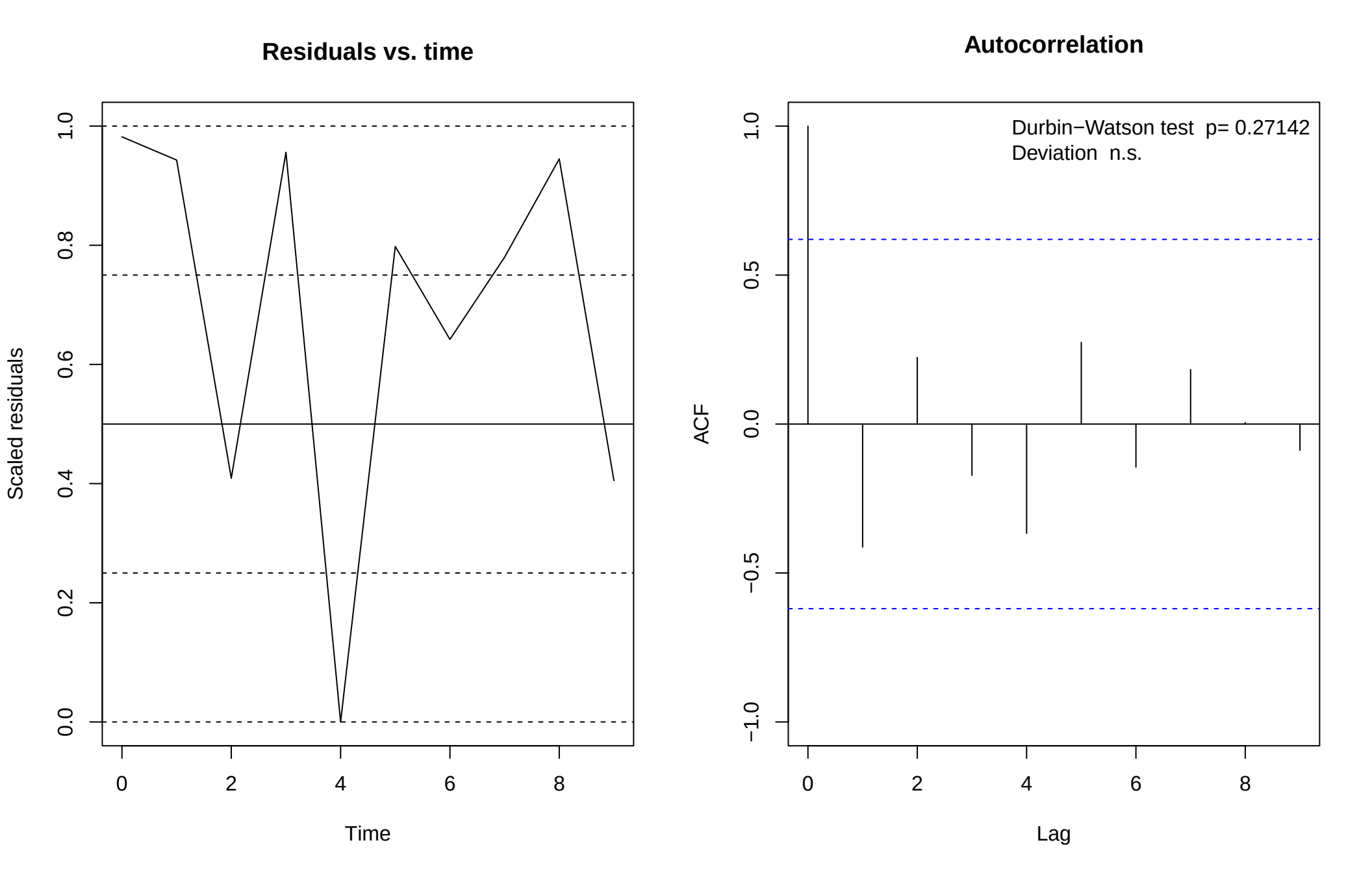

Residuals vs Distance_to_Clutter__m_
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Distance_to_Clutter__m_ (rank transformed)

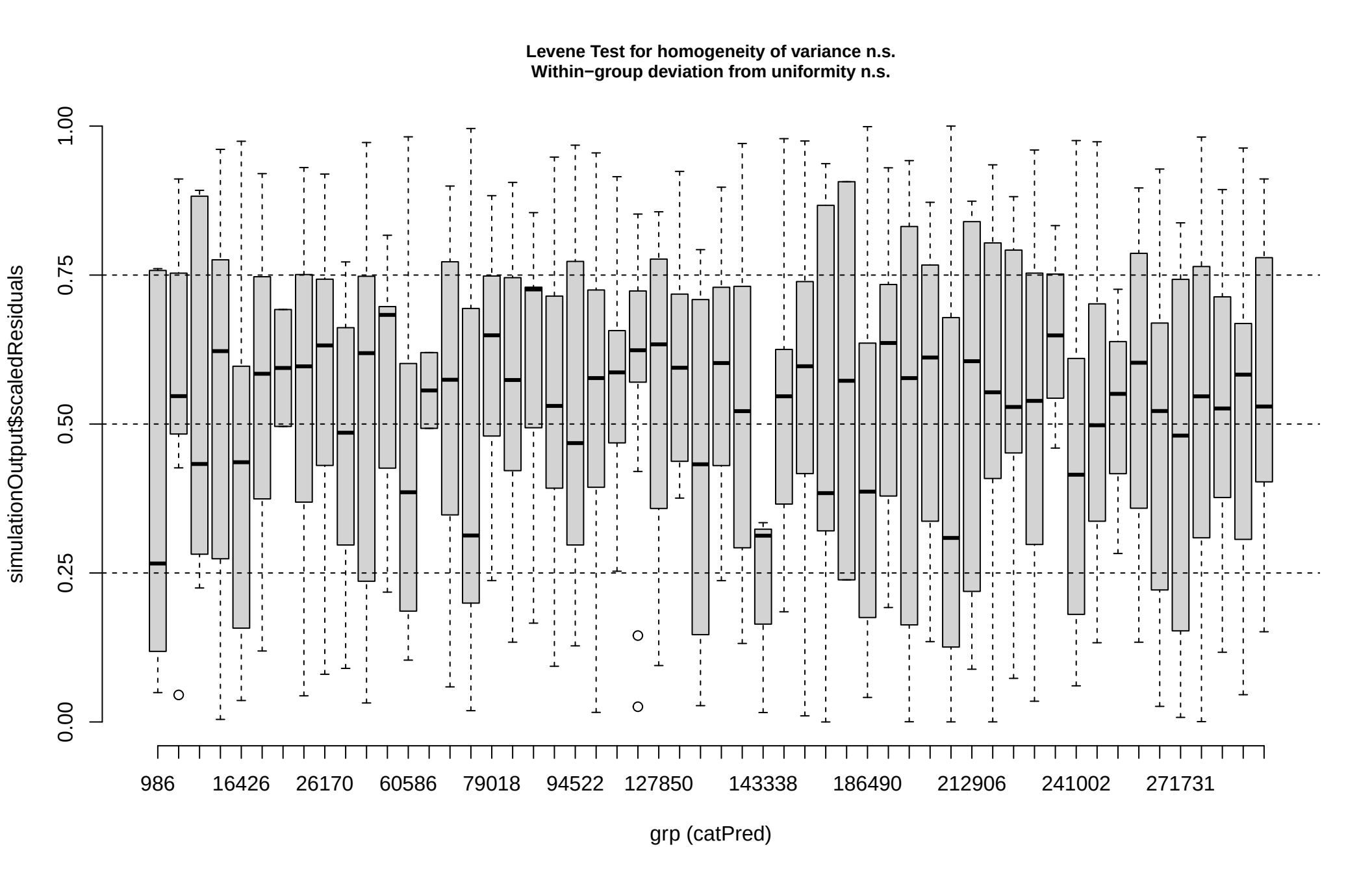

Levene Test for homogeneity of variance n.s.
Within−group deviations from uniformity significant (red)
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
0.381949458483755
1
Water_Nearby (rank transformed) (catPred)

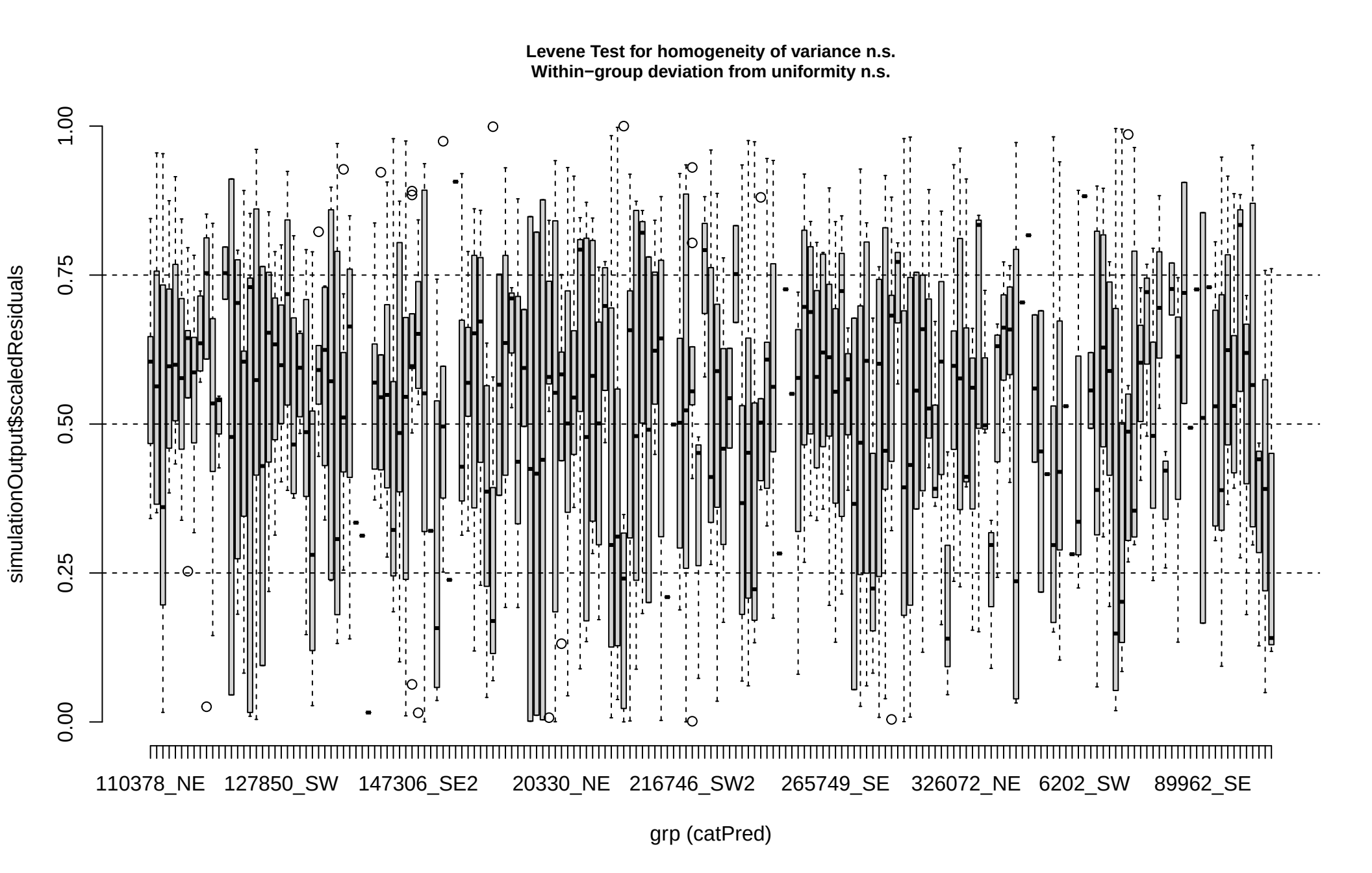

Residuals vs Nightly_Mean_Temp
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Temp (rank transformed)

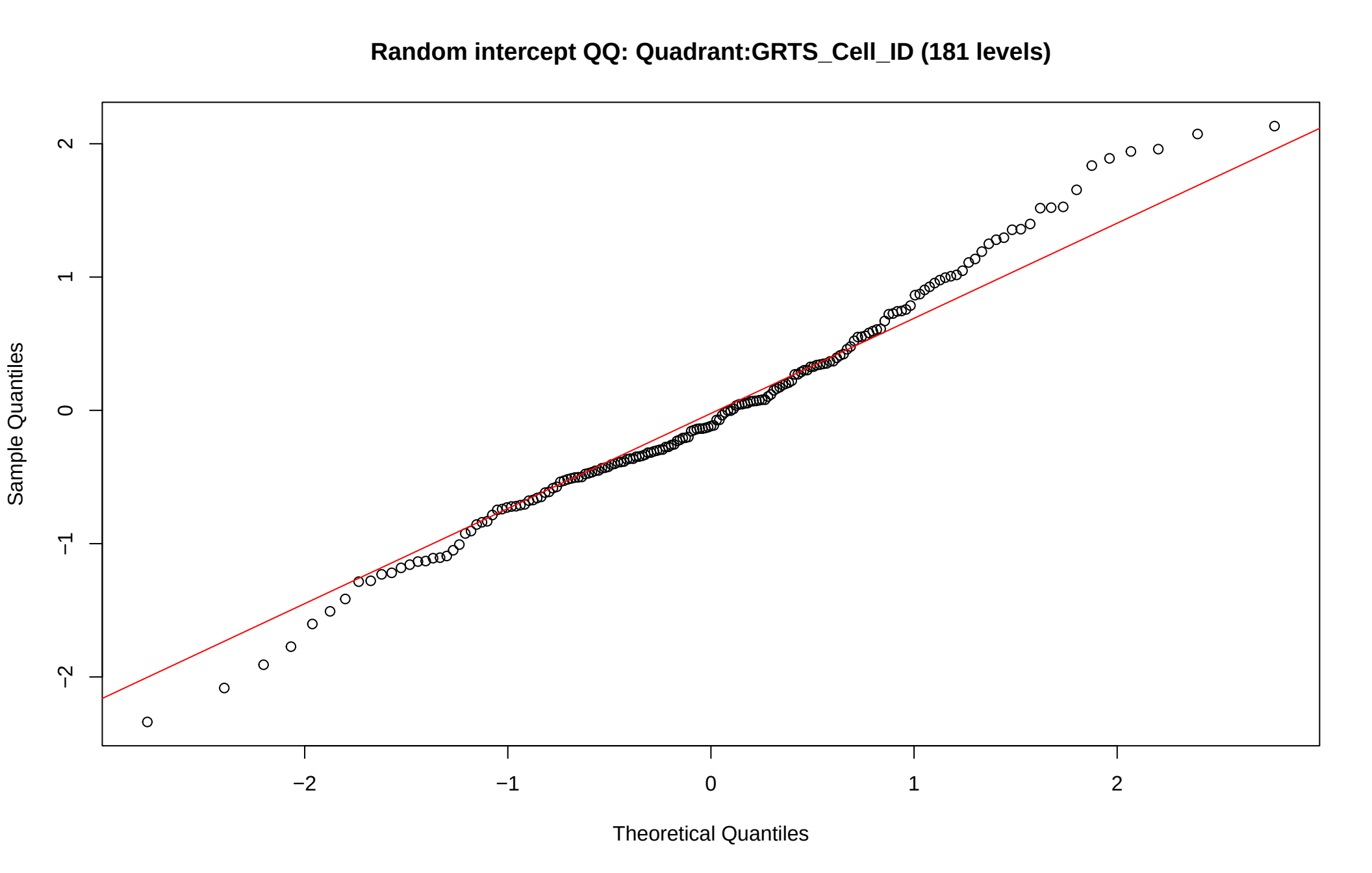

DHARMa nonparametric dispersion test via sd of
residuals fitted vs. simulated
Frequency
0
10
20
30
40
50
60
70
0
2
4
6
8
10
12
Simulated values, red line = fitted model. p−value (two.sided) = 0.958

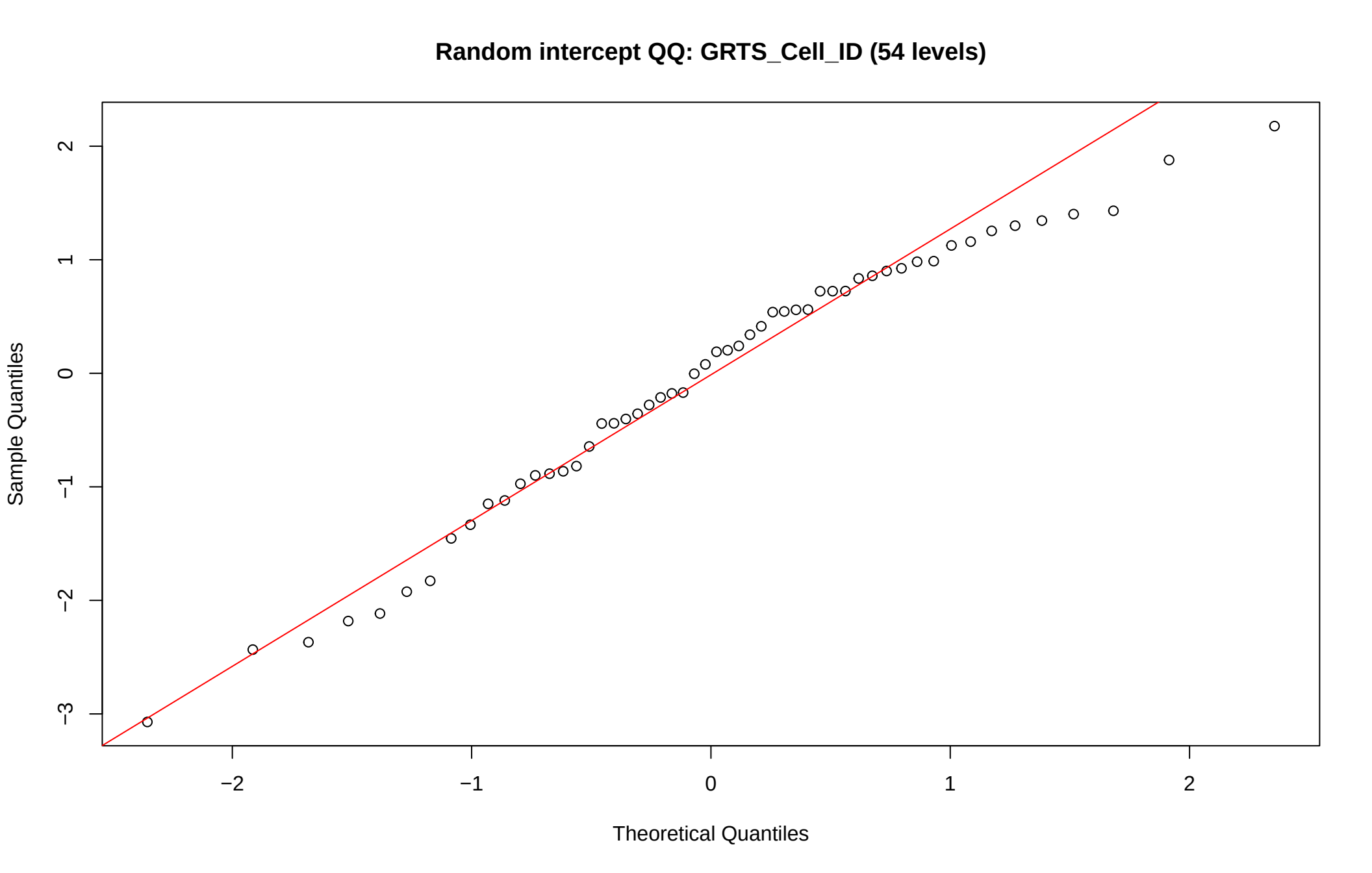

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
10
20
30
40
50
60
70
80
90
100
110
Simulated values, red line = fitted model. p−value (two.sided) = 0.244


Residuals vs. time
Scaled residuals
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Time
Autocorrelation
ACF
Durbin−Watson test p= 0.27142
Deviation n.s.
1.0
0.5
0.0
−0.5
−1.0
0
2
4
6
8
Lag


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
986
16426
26170
60586
79018
94522
127850
143338
186490
212906
241002
271731
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
110378_NE
127850_SW
147306_SE2
20330_NE
216746_SW2
265749_SE
326072_NE
6202_SW
89962_SE
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (181 levels)
Sample Quantiles
Theoretical Quantiles
−2
−1
0
1
2


Random intercept QQ: GRTS_Cell_ID (54 levels)
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
−2
−1
0
1
2